VALUATIVE CRITERION FOR PROJ OF A FINITELY GENERATED GRADED ALGEBRA

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Allowed standard prerequisites (editorial).

- Separatedness of Proj. lemma-proj-separated (carded this batch). Gives the uniqueness part of the criterion.
- Valuations. algebra-definition-value-group (cross-chapter, named). Supplies the value group Gamma and the valuation v of A.
- Coverage by standard opens. algebra-lemma-topology-proj (cross-chapter, named). The D_+(x_i) cover Proj(S), so some generator image is nonzero at the K-point.

Source reference. Original reference 00IE: focused-seeds/geometry/S9-cons-proj-valuative-criterion/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01MC: sources/stacks-constructions/pinned-original-source.tex, source lines 1532–1532. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01KD: focused-seeds/geometry/S9-cons-proj-valuative-criterion/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01KZ: focused-seeds/geometry/S9-cons-proj-valuative-criterion/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a graded ring. If S is finitely generated as an algebra over S_0, then the morphism $Proj(S) \to \mathrm{Spec}(S_0)$ satisfies the existence and uniqueness parts of the valuative criterion, see Schemes, Definition 01KD.*

Proof. The uniqueness part follows from the fact that $\mathrm{Proj}(S)$ is separated (Lemma 01MC and Schemes, Lemma 01KZ). Choose $x_i \in S_+$ homogeneous, $i = 1, \ldots, n$

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which generate S over S_0. Let $d_i = \deg(x_i)$ and set $d = \text{lcm}\{d_i\}$. Suppose we are given a diagram

$$\begin{array}{ccc} \text{Spec}(K) & \longrightarrow & \text{Proj}(S) \\ \downarrow & & \downarrow \\ \text{Spec}(A) & \longrightarrow & \text{Spec}(S_0) \end{array}$$

as in Schemes, Definition 01KD. Denote $v : K^* \to \Gamma$ the valuation of A, see Algebra, Definition 00IE. We may choose an $f \in S_+$ homogeneous such that $\text{Spec}(K)$ maps into $D_+(f)$. Then we get a commutative diagram of ring maps

$$\begin{array}{ccc} K & \xleftarrow{\varphi} & S_{(f)} \\ \uparrow & & \uparrow \\ A & \longleftarrow & S_0 \end{array}$$

After renumbering we may assume that $\varphi(x_i^{\deg(f)}/f^{d_i})$ is nonzero for $i = 1, \ldots, r$ and zero for $i = r+1, \ldots, n$. Since the open sets $D_+(x_i)$ cover $\text{Proj}(S)$ we see that $r \geq 1$. Let $i_0 \in \{1, \ldots, r\}$ be an index minimizing $\gamma_i = (d/d_i)v(\varphi(x_i^{\deg(f)}/f^{d_i}))$ in Γ. For convenience set $x_0 = x_{i_0}$ and $d_0 = d_{i_0}$. The ring map φ factors though a map $\varphi' : S_{(fx_0)} \to K$ which gives a ring map $S_{(x_0)} \to S_{(fx_0)} \to K$. The algebra $S_{(x_0)}$ is generated over S_0 by the elements $x_1^{e_1} \ldots x_n^{e_n}/x_0^{e_0}$, where $\sum e_i d_i = e_0 d_0$. If $e_i > 0$ for some $i > r$, then $\varphi'(x_1^{e_1} \ldots x_n^{e_n}/x_0^{e_0}) = 0$. If $e_i = 0$ for $i > r$, then we have

$$\begin{aligned} d\deg(f)v(\varphi'(x_1^{e_1} \ldots x_r^{e_r}/x_0^{e_0})) &= dv(\varphi'(x_1^{e_1 \deg(f)} \ldots x_r^{e_r \deg(f)}/x_0^{e_0 \deg(f)})) \\ &= d\sum e_i v(\varphi'(x_i^{\deg(f)}/f^{d_i})) - e_0 v(\varphi'(x_0^{\deg(f)}/f^{d_0})) \\ &= \sum e_i d_i \gamma_i - e_0 d_0 \gamma_0 \\ &\geq \sum e_i d_i \gamma_0 - e_0 d_0 \gamma_0 = 0 \end{aligned}$$

because γ_0 is minimal among the γ_i. This implies that $S_{(x_0)}$ maps into A via φ'. The corresponding morphism of schemes $\text{Spec}(A) \to \text{Spec}(S_{(x_0)}) = D_+(x_0) \subset \text{Proj}(S)$ provides the morphism fitting into the first commutative diagram of this proof. $\square$